VOLTRA E3 ELECTRIC BIKE - OWNER'S MANUAL

Welcome. This manual covers the Voltra E3 city e-bike, model year 2026. Read it fully before your first ride. Voltra Mobility Ltd. is headquartered in Pune, India, and has been building electric bicycles since 2019.

What is in the box: the bike with the front wheel removed, the 48 V battery pack, the 4 A smart charger, two keys for the battery lock, a 5 mm and 6 mm hex key, pedals, and this manual.

Key specifications. Motor: 250 W rear hub motor with a peak torque of 45 Nm. Battery: 48 V, 13 Ah lithium-ion pack (624 Wh). Top assisted speed: 25 km/h, which is the legal limit for pedal-assist bikes in India and the EU. Range: 60 to 90 km depending on assist level, rider weight and terrain. Frame: 6061 aluminium in sizes M (riders 160-175 cm) and L (riders 175-190 cm). Bike weight: 22.5 kg including the battery. Maximum total load: 120 kg including rider and cargo. Brakes: hydraulic disc brakes with 180 mm rotors. Tyres: 27.5 x 2.2 inch puncture-resistant tyres.

The display. The colour display on the left handlebar shows speed, battery level, assist level, trip distance and error codes. Press the power button for two seconds to switch the bike on or off. Use the plus and minus buttons to change the assist level. Hold the minus button for three seconds to switch the front light on.

ASSIST LEVELS AND RIDING

The E3 has five assist levels plus level 0, where the motor is off and the bike rides like a normal bicycle. Level 1 (Eco) gives the longest range and is best for flat city roads. Level 3 (Tour) is a balanced setting for daily commuting. Level 5 (Turbo) gives the most help on steep hills but drains the battery about three times faster than Eco.

The motor only helps while you pedal. A torque sensor in the bottom bracket measures how hard you press the pedals and adds power in proportion, so the ride feels natural. Assistance stops automatically at 25 km/h. You can ride faster under your own power.

Walk mode helps you push the bike up ramps or out of a basement parking. Hold the minus button for six seconds while standing next to the bike. The motor then moves the bike forward at up to 6 km/h for as long as you keep holding the button.

Riding in the rain is fine. The motor, battery and display are rated IP65 against dust and water jets. Do not submerge the bike, do not ride through water deeper than 15 cm, and never clean it with a pressure washer.

Before every ride, check that both brakes work, that the tyres are inflated, that the battery is locked in place and that the quick-release levers are closed.

BATTERY AND CHARGING

Charge the battery fully before the first ride. A full charge from empty takes about 4.5 hours with the included 4 A charger. Charging from 20 percent to 80 percent takes about 2 hours. The charger port is under a rubber cover on the left side of the battery.

You can charge the battery on the bike or remove it. To remove it, insert the key, turn it anticlockwise and lift the battery out of the down tube. Charge indoors at a temperature between 10 and 30 degrees Celsius. Never charge a battery that is below 0 degrees; let it warm up to room temperature first.

The battery has a battery management system (BMS) that protects it from overcharging, deep discharge and overheating. The LED on the charger is red while charging and turns green when the battery is full.

Battery care. Lithium-ion batteries last longest when kept between 20 and 80 percent. If you will not ride for more than a month, store the battery at about 60 percent charge in a cool, dry place and top it up every three months. The battery is designed for about 800 full charge cycles, after which it still holds at least 70 percent of its original capacity.

Only use the original Voltra charger. Using a third-party charger can damage the cells and voids the battery warranty.

ERROR CODES

When the system detects a problem, the display shows an error code and the motor stops helping. Most errors clear after switching the bike off and on.

E-01: Throttle or button fault. Check that no button is stuck and restart the bike. E-07: Battery voltage too low. Charge the battery. E-12: Motor overheating. This can happen on long, steep climbs in Turbo mode. Stop for 15 minutes or switch to a lower assist level; assistance returns when the motor cools down. E-21: Speed sensor fault. The magnet on the rear wheel spoke may have moved. Align it with the sensor on the frame so the gap is between 2 and 5 mm. E-30: Communication error between the display and the controller. Check that the display cable connector is fully pushed in. E-44: Battery temperature out of range. Move the bike to a place between 0 and 40 degrees Celsius and wait. E-52: Torque sensor fault. Do not ride with assistance. Contact an authorised service centre.

If an error code keeps coming back after a restart, note the code and contact Voltra support at support@voltra.example or call 1800-555-0136 (toll free, Monday to Saturday, 9 am to 6 pm IST).

MAINTENANCE

Tyre pressure: keep the tyres between 40 and 55 psi. Use the lower value for comfort on rough roads and the higher value for range on smooth roads. Check pressure every week; low pressure is the most common cause of reduced range.

Chain: clean and lubricate the chain every 300 km, or after every wet ride. Wipe off old oil with a dry cloth before applying new chain lubricant.

Brakes: the hydraulic brakes adjust themselves as the pads wear. Replace the brake pads when they are thinner than 1.5 mm. A squealing noise usually means oil or dirt on the rotor; clean the rotor with isopropyl alcohol.

Service schedule: the first free service is due after 500 km or 3 months, whichever comes first. After that, have the bike serviced every 2,000 km or once a year. Services at authorised centres include a firmware update for the motor controller.

Torque values for bolts: stem bolts 6 Nm, handlebar clamp 5 Nm, seat post clamp 7 Nm, pedals 35 Nm. Over-tightening can crack the aluminium frame parts.

Software updates can also be installed from the Voltra Ride mobile app over Bluetooth. Keep the bike switched on and close to the phone during the update, which takes about 10 minutes.

WARRANTY AND SUPPORT

Warranty: the frame is covered for 5 years against manufacturing defects. The motor and the battery are covered for 2 years or 20,000 km, whichever comes first. The display, charger and other electrical parts are covered for 1 year. Wear parts such as tyres, brake pads, chain and grips are not covered.

The warranty does not cover damage from accidents, misuse, racing, water submersion, unauthorised modifications such as speed unlocking kits, or use of a non-original charger. To claim warranty, you need the original invoice and the frame number, which is stamped under the bottom bracket.

Returns: you can return an unused bike within 14 days of delivery for a full refund. A ridden bike can be returned within 14 days with a 10 percent restocking fee, as long as it has less than 50 km on the odometer.

Theft protection: register your frame number in the Voltra Ride app. The app can lock the motor remotely, so a stolen bike will not give any assistance. Voltra does not offer theft insurance; ask your home insurance provider about cover for e-bikes.

Disposal: never put the battery in household waste. Return old batteries to any Voltra dealer, who will recycle them free of charge.